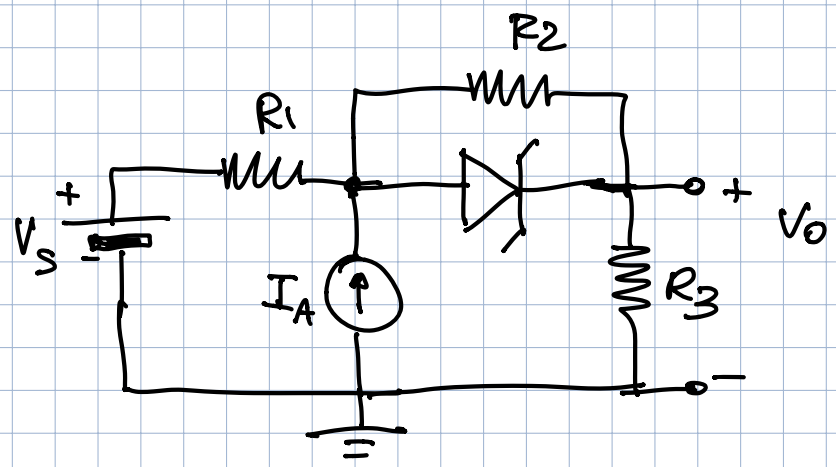


$R_1 = 1\,K\Omega$ $\quad I_A = 10\,mA$

$R_2 = 2\,K\Omega$ $\quad D_Z$: $V_{ON} = 1V$

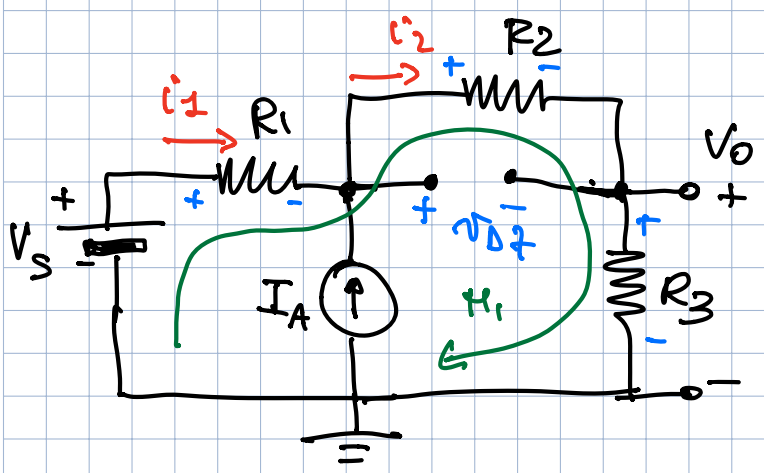
$R_{ON} = 0\,\Omega$

$R_3 = 3\,K\Omega$ $\quad V_Z = 10V$

$R_Z = 0\,\Omega$

trovare $V_O = f(V_S)$

Ip. $D_Z = OFF$



M_1: $-V_S + R_1 i_1 + R_2 i_2 + R_3 i_2 = 0$

$i_2 = i_1 + I_A$

$-V_S + R_1 i_1 + (R_2 + R_3) i_1 + (R_2 + R_3) I_A = 0$

$$i_1 = \frac{V_S - (R_2 + R_3) I_A}{R_1 + R_2 + R_3}$$

$$i_2 = i_1 + I_A = \frac{V_S + R_1 I_A}{R_1 + R_2 + R_3}$$

$$v_{DZ} = i_2 R_2 = \frac{R_2}{R_1 + R_2 + R_3} \cdot (V_S + R_1 I_A)$$

$\boxed{v_{DZ} < -V_Z}$

$\Rightarrow D_Z$ = BREAK ZENER

$$\frac{R_2}{R_1 + R_2 + R_3}(V_S + R_1 I_A) < -V_Z$$

$$V_S < -V_Z \frac{R_1 + R_2 + R_3}{R_2} - R_1 I_A$$

$$V_S < -10V \cdot \frac{6}{2} - 10V$$

$\boxed{V_S < -40V}$ $\Rightarrow$ DZ in BREAK

$\boxed{v_{DZ} > V_{ON}}$

$\Rightarrow D_Z$ = "ON"

$$\frac{R_2}{R_1 + R_2 + R_3}(V_S + R_1 I_A) > V_{ON}$$

$$V_S > V_{ON} \frac{R_1 + R_2 + R_3}{R_2} - R_1 I_A$$

$$V_S > 1V \cdot \frac{6}{2} - 10V$$

$\boxed{V_S > -7\,V}$ $\Rightarrow$ DZ "ON"

$\Rightarrow \boxed{-40V \le V_S \le -7V} \Rightarrow D_Z = \text{"OFF"}$

1) PER $-40V \le V_S \le -7V \Rightarrow$ DIODO OFF

(VEDI CIRCUITO SOPRA)

$$v_O = R_3 i_2 = \frac{R_3}{R_1+R_2+R_3}(V_S + R_1 I_A) = \frac{3}{6}V_S + \frac{3}{6}\cdot 10V$$

$$\Rightarrow \boxed{v_O = \frac{V_S}{2} + 5V}$$

RACCORDI

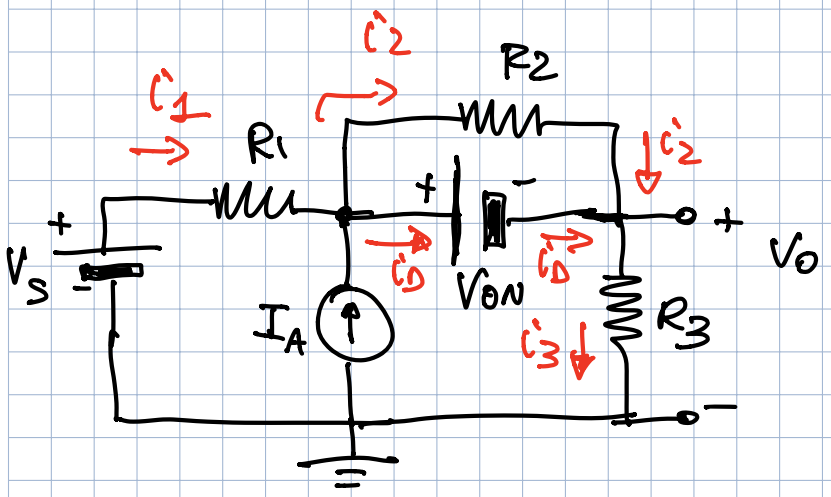
$$V_O(-7) = -\frac{7}{2} + 5V = 1{,}5V$$

$$V_O(-40V) = -\frac{40}{2} + 5 = -15V$$

$$m_1 = 0 \quad q = 5V$$

VERIFICARE CHE $-V_Z \le v_{DZ} \le V_{ON}$

2) PER $V_S > -7V \Rightarrow$ DIODO "ON"



$$i_2 = \frac{V_{ON}}{R_2} = \frac{1V}{2K\Omega} = 0{,}5mA$$

$$i_1 + I_A = i_2 + i_D$$

$$\Rightarrow i_D = i_1 - i_2 + I_A$$

$$i_3 = i_2 + i_D = i_1 + I_A$$

$$-V_S + i_1 R_1 + V_{ON} + R_3 i_3 = 0$$

$$-V_S + i_1 R_1 + V_{ON} + R_3 i_1 + R_3 I_A = 0$$

$$i_1 = \frac{V_S - V_{ON} - R_3 I_A}{R_1 + R_3} = \frac{V_S}{R_1+R_3} - \frac{V_{ON} + R_3 I_A}{R_1+R_3}$$

$$i_3 = i_1 + I_A = \frac{V_S}{R_1+R_3} - \frac{V_{ON}}{R_1+R_3} - \frac{R_3 I_A}{R_1+R_3} + I_A$$

$$v_O = i_3 R_3 = \frac{R_3}{R_1+R_3} V_S - V_{ON}\frac{R_3}{R_1+R_3} + \frac{R_1 I_A}{R_1+R_3}\cdot R_3$$

$$v_O = \frac{3}{4} V_S - \frac{3}{4}\cdot 1V + \frac{3}{4}\cdot 10V$$

$$\Rightarrow \boxed{v_O = \frac{3}{4} v_S + \frac{27}{4} V}$$

$m = 0{,}75 \qquad q = 6{,}75V$

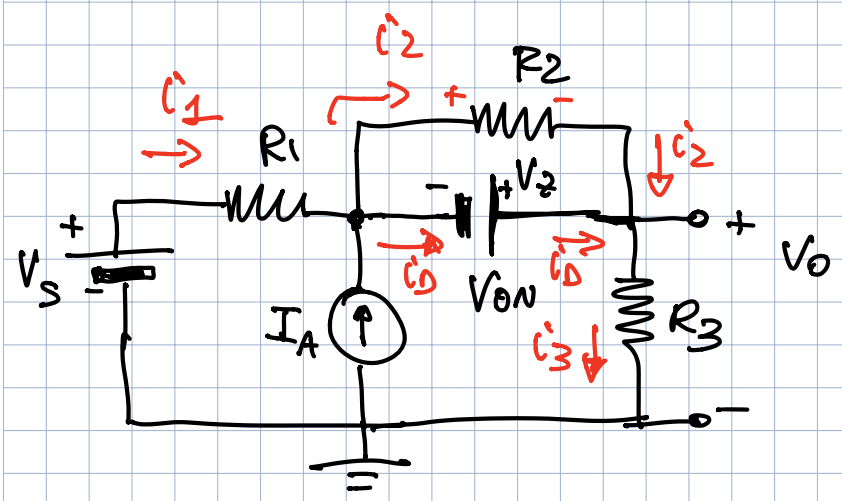
VERIFICARE CHE $i_D > 0$

RACCORDO:

$$v_O(-7V) = \frac{3}{4}\cdot(-7V) + \frac{27}{4}V = -\frac{21}{4}V + \frac{27}{4}V = \frac{6}{4}V = \frac{3}{2}V = 1{,}5V$$

OK RACCORDO

PER $V_S < -40V \Rightarrow$ D_Z IN BREAKDOWN



$$i_2 = -\frac{V_Z}{R_2}$$

$$i_1 + I_A = i_2 + i_D$$

$$\Rightarrow i_D = i_1 - i_2 + I_A$$

$$i_3 = i_2 + i_D = i_1 + I_A$$

$$-V_S + i_1 R_1 - V_Z + R_3 i_3 = 0$$

$$-V_S + i_1 R_1 - V_Z + R_3 i_1 + R_3 I_A = 0$$

$$i_1 = \frac{V_S + V_Z - R_3 I_A}{R_1 + R_3} = \frac{V_S}{R_1+R_3} + \frac{V_Z - R_3 I_A}{R_1+R_3}$$

$$i_3 = i_1 + I_A = \frac{V_S}{R_1+R_3} + \frac{V_Z}{R_1+R_3} - \frac{R_3 I_A}{R_1+R_3} + I_A$$

$$v_0 = i_3 R_3 = \frac{R_3}{R_1+R_3} V_S + V_Z \frac{R_3}{R_1+R_3} + \frac{R_1 I_A}{R_1+R_3} \cdot R_3$$

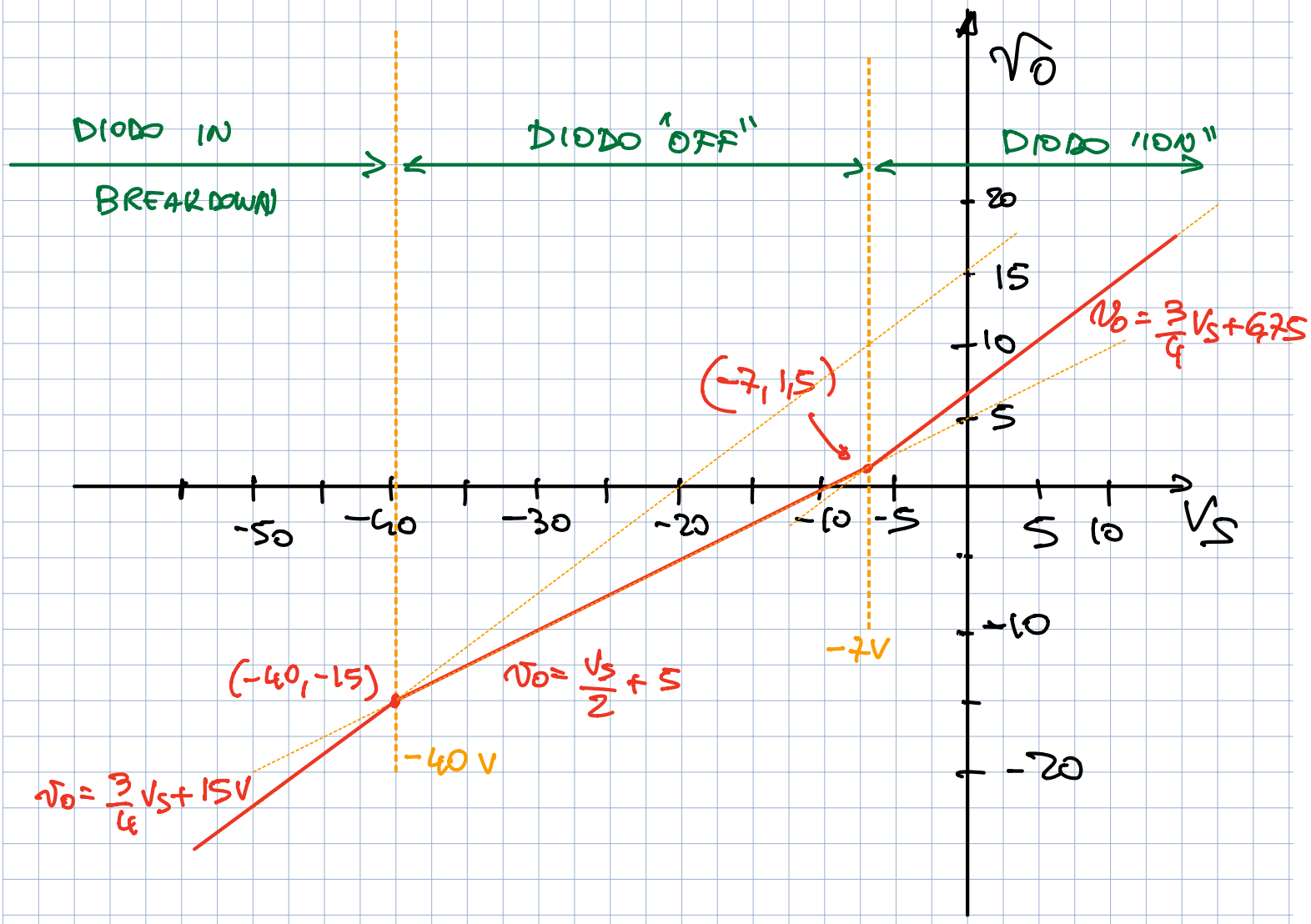
$$v_0 = \frac{3}{4} V_S + \frac{3}{4} \cdot 10V + \frac{3}{4} \cdot 10V$$

$$= \boxed{\frac{3}{4} V_S + 15V}$$

$m = 0{,}75 \quad q = 15V$

RACCORDO

$$V_0(-40V) = \frac{3}{4}(-40V) + 15V$$

$$= -15V$$

OK RACCORDO